Eberswalde University for Sustainable Development

FIT / FST / GCM 1st term – winter term 2025/26

Process Modelling Methodology (PMM)

Examination task (announced on January 23th 2026)

Submission form:	**term paper (minimum: 6 pages) uploaded as PDF via submission folder on Moodle**
Submission deadline:	**February 22st 2026 at 11.59 p.m. CET via Moodle**

Goal:

The goal of this term paper is

- to compare the growth of observed *Picea abies* trees on two different locations.
- to select and fit appropriate forest growth models to visualize tree growth of *Picea abies* at both locations and compare it.
- to present the fitted forest growth models with respect to their coefficients.
- to select a suitable allometric model to calculate the above-ground biomass stored in a representative 120-year-old tree of the species *Picea abies* at the respective locations and compare it (it is acceptable to consider the average situation).
- to reflect and describe precisely the reasons for the selecting the applied allometric model(s) (keywords: range, sample size, coefficient of determination).
- to present the respective results.
- to interpret and discuss the results with respect to differences between the two locations (e.g. composition of tree stands, climatic conditions, elevation etc.)
- to create a document (pdf) for your final submission (including tables and figures and following the scientific writing principles described at the end).

Data & Software:

You are free to use any appropriate software.
The data show the results of individual tree observations over time in two different *Picea abies* tree stands.

The attributes provided in the data file are:

location =	1 = cold temperate climate, altitude > 1000 m above sea level; 2 = temperate climate, altitude < 1000 m above sea level (e.g. Harz, Germany)
tree =	ID of observed *Picea abies* trees

age.base =	Age of respective tree
height =	Measured stem height of respective tree and age [m]
dbh.cm =	Measured stem diameter at breast height per tree and age [cm]
volume =	Calculated stem volume per tree and age [m^3]

Hints:

- In case you want to export a figure from RStudio use the button 'Export' of the bottom right window after plotting and follow the export assistant.
- In case you want to subset the data use subset() in RStudio; run *?subset* for help.
- Do not forget to install and activate the required packages!
- Do not forget to tick *Strings as factors* during data import or to define factor levels, when necessary, after data import (cf. as.factor())

Remark:

It is not necessary to document every processing step of the data and / or your RStudio script when working through the task. You may name and cite the respective software and / or packages. Concentrate on explaining and introducing the selected approach and most important settings, on the results, their presentation (e.g. tables, figures) and interpretation. Do not forget to give references and transparently note when you used further tools.

Good luck & much success ☺

Guidance for scientific writing:

1) General Formalities

- **Correct orthography and punctuation** have to be used; numerous spelling mistakes and unclear sentences lead to downgrading of the work.
- A **factual style** and no colloquial language are to be used.
- An electronic version as a pdf file has to be submitted **to the respective supervisor / lecturer the latest at the official deadline (end of the examination period).**
- **Figures and tables** have to be included in the text close to the corresponding passage. It is important to note that they can only support an argumentation, not replace it, i.e. an uncommented figure is rather detrimental than helpful. Figures and tables have to be titled and serially numbered. Figure captions and table headings are used! A list of figures and tables is not necessary.

2) Structure of the Report

The report should have the following **structure**:

- Title page
- Text: The text should follow a **general IMRaD structure** (Introduction, Material & Methods, Results & Discussion). The introduction should stress the significance of the question, establish a classification or demarcation of the topic and present a synopsis of the thoughts developed in the following chapters.
- Reference list: The reference list has to be complete and the **common APA citation style** should be applied (Inform yourself here: https://www.scribbr.com/category/citing-sources/). Any type of plagiarism needs to be avoided and might lead to downgrading of the work!

  "Plagiarism is presenting someone else's work or ideas as your own, with or without their consent, by incorporating it into your work without full acknowledgement. All published and unpublished material, whether in manuscript, printed or electronic form, is covered under this definition. Plagiarism may be intentional or reckless, or unintentional. Under the regulations for examinations, intentional or reckless plagiarism is a disciplinary offence."
  Oxford University. (2021, September 24). Plagiarism. Oxford University. https://www.ox.ac.uk/students/academic/guidance/skills/plagiarism

3) Formatting of the term paper

Authors are free to define the format of their written work independently. In general, the following rules should be respected:

- Page set up (e.g. margins, page numbers) and text format (e.g. line spacing, font type text body, font type headlines, font type figure captions and table headings) should be defined once and applied **consistently** throughout the text! It is recommended to use at least 1.3 rows as line spacing and page margins of at least 2.5 cm (top, right, left) and 2.0 cm (bottom).
- In addition, the structure has to be **consistent**, i.e. each level has to include at least two subordinate levels (e.g., '2. section title' must be followed by at least '2.1 subsection title one' and '2.2. subsection title two'). As a rule of thumb, the text under each level of structuring should be at least half a page long, but should preferably not exceed three pages.